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The EUV Snapshot Imaging Spectrograph

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1. INTRODUCTION

The light emitted by the solar transition region (TR) and corona varies significantly as a function of position, wavelength, and time. When viewed from Earth, the spectral radiance from the Sun can be written as: $I(x, y, \lambda, t)$, where x and y are the helioprojective Cartesian coordinates (W. T. Thompson 2006), λ is wavelength, and t is time. The ideal solar imaging spectrograph would capture $I(x, y, \lambda, t)$ with high resolution in x, y, λ, and t *and* over a wide field of view (FOV), wavelength range, and time period. Of course, the temporal dimension is privileged, so we often reduce the problem to capturing a 3D spatial/spectral cube at a particular time t_0: $I(x, y, \lambda, t_0)$. Since we use 2D detectors, this means that we must find a way to flatten the 3D cube into two dimensions without losing information.

One obvious way to accomplish this is to multiplex one of the three remaining dimensions in time. Narrowband, tunable filters, such as the GREGOR Fabry–Pérot Interferometer (K. G. Puschmann et al. 2012) or CRISP at the Swedish Solar Telescope (G. B. Scharmer et al. 2008), multiplex the wavelength dimension in time, and can change the selected wavelength in 50 ms or less (G. B. Scharmer et al. 2008), but the technology does not exist to use this technique for wavelengths shorter than ∼150 nm (J.-P. Wuelser et al. 2000). The nearest extreme ultraviolet (EUV) equivalent is a multilayer-coated imager such as the Transition Region and Coronal Explorer (TRACE) (B. N. Handy et al. 1999) or the Atmospheric Imaging Assembly (AIA) (J. R. Lemen et al. 2012), which abandons the wavelength dimension entirely and integrates over a passband holding many emission lines. Two such passbands can be compared to infer a Doppler shift (T. Sakao et al. 1999), but weaker lines within them bound the velocity resolution near $\sim 1000\,\mathrm{km\,s^{-1}}$ (K. Kobayashi et al. 2000), coarser than the flows that characterize TR dynamics.

A spatial dimension can be multiplexed instead. An entrance slit admits a single column of the scene, so the detector records λ against y without ambiguity, and x is recovered by stepping the slit across the target. The Interface Region Imaging Spectrograph (IRIS) (B. De Pontieu et al. 2014) and the Spectral Imaging of the Coronal Environment (SPICE) aboard Solar Orbiter (SPICE Consortium et al. 2020) work this way, and the spectra they return are the most faithful of any technique discussed here. What is sacrificed is simultaneity: neighboring columns of the reconstructed cube are separated in time by the raster, and structure that evolves faster than the raster completes is smeared along x. The Multi-slit Solar Explorer (MUSE) (B. De Pontieu et al. 2020, 2022; M. C. M. Cheung et al. 2022) narrows this gap considerably by ruling 37 slits across the field and separating their overlapping spectra afterward, which shortens the raster of an active region to as little as 12 s.

A third strategy divides the detector rather than the exposure. An integral field spectrograph slices the field of view and reformats those slices so that each is dispersed onto its own region of the detector, so the entire cube arrives in one exposure, at the cost of trading field of view against wavelength range. The microlensed hyperspectral imager of M. van Noort et al. (2022) demonstrates the technique at visible wavelengths, and the Solar eruptioN Integral Field Spectrograph (SNIFS) (V. L. Herde et al. 2024), flown in 2025, demonstrates it in the far ultraviolet (FUV). Extending it to the EUV remains out of reach of available optics.

The last strategy is to allow the flattening to be ambiguous and to undo it afterward. A spectrograph with no entrance slit disperses the whole field at once, so every emission line forms an image of the Sun and

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those images superimpose on the detector. The resulting frame, an *overlappogram*, encodes x and λ along a single axis, and the two cannot be separated within one exposure. Overlappograms are as old as spaceborne solar spectroscopy, having been recorded by the Naval Research Laboratory's (NRL's) S082A spectroheliograph (R. Tousey et al. 1973, 1977), which yielded both a census of EUV transitions (U. Feldman et al. 1985) and, much later, flare line ratios (F. P. Keenan et al. 2006); the technique is in use today at soft X-ray wavelengths by the Marshall Grazing Incidence X-ray Spectrometer (MaGIXS) (S. L. Savage et al. 2023). What has changed since S082A is that the ambiguity has become tractable. Reconstructing the cube from superimposed images is a tomographic problem (A. C. Kak & M. Slaney 1988), in which each diffraction order views the cube from a different angle and the number of independent views limits how much of the line profile can be recovered (M. R. Descour et al. 1997). Computed tomography imaging spectrographs (CTISs) (T. Okamoto & I. Yamaguchi 1991; F. V. Bulygin et al. 1991; M. Descour & E. Dereniak 1995) carry this to its extreme, projecting as many as 25 orders onto a single detector, with computational cost rising accordingly (N. Hagen & E. L. Dereniak 2008; N. Hagen & M. W. Kudenov 2013). Fewer views can suffice when the scene is sparse: C. DeForest et al. (2004) synthesized a magnetogram from a single exposure using only two dispersed orders. Inversion methods for overlapping spectral images have advanced considerably since (A. R. Winebarger et al. 2019; J. M. Davila et al. 2019; U. Kamaci & F. Kamalabadi 2026).

The Multi-Order Solar EUV Spectrograph (MOSES) (J. L. Fox et al. 2010; J. L. Fox 2011) brought this strategy to the solar EUV. A single concave grating forms three images at once, the undispersed $m = 0$ order and the $m = \pm 1$ orders, on three detectors, while a multilayer coating narrows the passband to a few emission lines so that the cube to be recovered is sparse enough to invert. Two flights showed that the concept works. J. L. Fox et al. (2010) measured the Doppler shift and width of explosive event line profiles in the TR; T. Rust & C. C. Kankelborg (2019) resolved bimodal profiles in quiet Sun events; H. T. Courrier & C. C. Kankelborg (2018) recovered Doppler maps using local correlation tracking; and J. D. Parker & C. C. Kankelborg (2022) separated the contributions of individual spectral lines. The same flights exposed where the design binds. Only three views are available, one of which is undispersed and therefore contributes no spectral information, which caps the number of degrees of freedom that an inversion can recover; and because the dispersed orders lie in a plane, they fill the cylindrical volume of a sounding rocket payload poorly.

The EUV Snapshot Imaging Spectrograph (ESIS) was designed to relax both constraints. An array of gratings arranged about the optical axis re-images the prime focus of a single primary mirror, each dispersing at a different angle, so that every detector records a dispersed view and the views are distributed around the payload rather than confined to a plane. ESIS flew for the first time on 2019 September 30, and first results from that flight are reported by J. D. Parker et al. (2022). An earlier form of the design was described by H. T. Courrier (2020); the instrument that flew differs from it, most consequentially in the removal of the primary mask, which admits vignetting into the system. This paper describes the instrument as it was built and flown. In Section 2 we explain how experience with MOSES shaped the design of ESIS. Section 3 states the scientific objectives and the requirements they impose. Section **??** describes the instrument itself, and Section **??** the mission profile.

2. THE ESIS CONCEPT

A primary goal of the ESIS instrument is to improve upon the imaging spectroscopy demonstrated by MOSES. Therefore, the design of the new instrument draws heavily from experiences and lessons learned through two flights of MOSES. ESIS and MOSES are both CTIS instruments. As such, both produce overlappograms of a narrow portion of the solar spectrum, with the goal of enabling the reconstruction of a spectral line profile at every point in the field of view. The similarities end there, however, as the optical layout of ESIS differs significantly from that of MOSES. In this section, we detail some difficulties and limitations encountered with MOSES, then describe how the new design of ESIS addresses these issues.

2.1. *Limitations of the MOSES Design*

The MOSES design features a single concave diffraction grating forming images on three charge-coupled device (CCD) detectors (J. L. Fox et al. 2010) (Figure 1). The optical path is folded in half by a single flat secondary mirror (omitted in Figure 1). Provided that the three cameras are positioned correctly, this arrangement allows the entire telescope to be brought into focus using only the central (undispersed) order and a visible light source. Unfortunately this design uses volume inefficiently for two reasons. First, the lack of magnification by the secondary mirror limits the folded length of the entire telescope to be no less than half of the 5 m focal length of the grating (J. L. Fox et al. 2010; J. L. Fox 2011). Second, the resolving power of the instrument is limited by the ratio of camera separation to pixel size. To achieve the maximum dispersion of 29 km pix^{-1} s^{-1} (J. L. Fox et al. 2010), the outboard orders are imaged as far apart as possible in the ~ 0.6 m diameter envelope of the rocket payload. This planar arrangement leaves much unused space in the cylindrical volume of the payload.

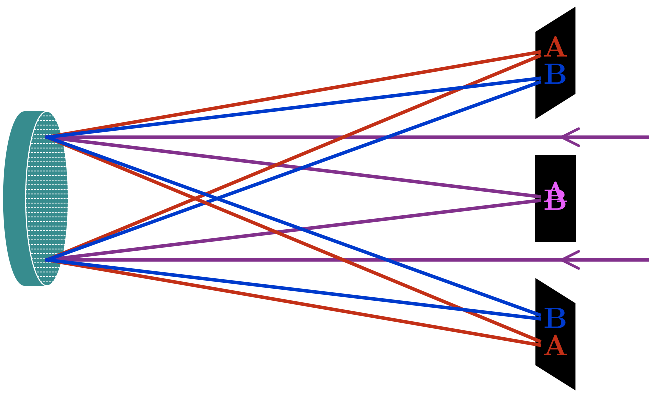

Figure 1. Schematic diagram of the MOSES instrument. Incident light from the right forms an undispersed image on the central $m = 0$ CCD. Dispersed images are formed on the outboard $m = \pm 1$ CCDs. The imaging of longer and shorter wavelengths is indicated by the use of red and blue letters respectively.

Furthermore, the monolithic secondary, though it confers the focus advantage noted above, does not allow efficient placement of the $m = \pm 1$ CCDs. For all practical purposes, the payload can only accommodate three diffraction orders ($m = -1, 0, +1$). Therefore, *MOSES can only collect, at most, three pieces of information at each point in the field of view.* From this, it is not reasonable to expect the reconstruction of more than three degrees of freedom for each spectral line, except in the case of very compact, isolated features such as those described by J. L. Fox et al. (2010) and T. Rust & C. C. Kankelborg (2019). Consequently, it is a reasonable approximation to say that MOSES is sensitive primarily to spectral line intensities, shifts, and widths (C. C. Kankelborg & R. J. Thomas 2001). With any tomographic apparatus, the degree of detail that can be resolved in the object depends critically on the number of viewing angles (A. C. Kak & M. Slaney 1988; M. R. Descour et al. 1997; N. Hagen & E. L. Dereniak 2008). So it is with the spectrum we observe with MOSES: more dispersed images are required to confer sensitivity to finer spectral details such as higher moments of the spectral line shape.

A related issue stems from the use of a single dispersion plane. Since the solar corona and transition region are structured by magnetic fields, the scene tends to be dominated by field-aligned structures such as loops (R. Rosner et al. 1978; R. M. Bonnet et al. 1980). When the MOSES dispersion direction happens to be aligned nearly perpendicular to the magnetic field, filamentary structures on the transition region serve almost as spectrograph slits unto themselves. The estimation of Doppler shifts then becomes a simple act of triangulation, and broadenings are also readily diagnosed (J. L. Fox et al. 2010; H. T. Courrier & C. C. Kankelborg 2018). A double-peaked profile can also be observed with sufficiently isolated features (T. Rust & C. C. Kankelborg 2019). Unfortunately, solar magnetic fields in the transition region are quite complex and do not have a global preferred direction. In cases where the field is nearly parallel to the instrument dispersion, spectral shifts and broadenings are not readily apparent.

The single diffraction grating also leads to a compromise in the optical performance of the instrument. Since the MOSES grating forms images in three orders simultaneously, there are not enough degrees of freedom in the optical system to control aberrations in all three spectral orders. During the first mission, MOSES was flown with a small amount of defocus (T. Rust & C. C. Kankelborg 2019), which exacerbated the inter-order point-spread function (PSF) variation and caused the individual PSFs to span several pixels (T. Rust & C. C. Kankelborg 2019; S. Atwood & C. Kankelborg 2018). The outboard PSFs were elongated, with their major axes at different angles. This resulted in spurious spectral features that require additional consideration (S. Atwood & C. Kankelborg 2018) and further increase the complexity of the inversion process (T. Rust & C. C. Kankelborg 2019; H. T. Courrier & C. C. Kankelborg 2018).

As MOSES lacks a field stop, it is possible for the grating to image off-band radiation on the detector from solar features that lie outside the intended FOV. Although the intensity of this off-band radiation is relatively low, it is not negligible. J. D. Parker & C. C. Kankelborg (2022) compared synthetic MOSES images to the real data and found that approximately ten percent of the intensity in the zeroth order image originated from more than ten dim lines in the MOSES passband, most of them too faint to be visible in the dispersed images. This study revealed that an undispersed ($m = 0$) channel, although attractive due to its lack of spatial-spectral ambiguity, is of limited utility due to spectral contamination. Moreover, the FOV should be clearly defined, and the same, for each image, so that the spectral and spatial content of each image is unambiguous.

Finally, the exposure cadence of MOSES is hindered by a $\sim$6 s readout time for the CCDs (J. L. Fox 2011). The observing interval for a solar sounding rocket flight is typically about five minutes. Consequently, every second of observing time is precious, both to achieve adequate exposure time and to catch the full development of dynamical phenomena. The MOSES observing duty cycle is ∼50 % since it is limited by the readout time of its CCDs. Thus, valuable observing time is lost. The readout data gap compelled us to develop a sequence with exposures ranging from 0.25-24 s, a careful trade-off between deep and fast exposures.

In summary, our experience leads us to conclude that the MOSES design has the following primary limitations:

1. inefficient use of volume
2. dispersion constrained by payload dimensions
3. too few dispersed images (orders)
4. single dispersion plane
5. lack of aberration control
6. poorly-defined and wavelength dependent FOV
7. spectral contamination
8. low duty cycle

In designing ESIS, we have sought to improve upon each of these points.

2.2. *ESIS Features*

The layout of ESIS (Figure 2) is a modified form of Gregorian telescope. Incoming light is brought to focus at an octagonal field stop by a parabolic primary mirror. In the ESIS layout, the secondary mirror of a typical Gregorian telescope is replaced by a segmented, octagonal array of diffraction gratings. From the field stop, the gratings re-image to CCD detectors arranged radially around the primary mirror. The gratings are blazed for first order, so that each CCD is fed by a single corresponding grating, and all the gratings are identical in design. The features of this new layout address all of the limitations described in Section 2.1, and are summarized here.

Replacing the secondary mirror with an array of concave diffraction gratings confers several advantages to ESIS over MOSES. First, the concavity of the gratings creates magnification in the ESIS optical system, which results in a shorter axial length than MOSES, without sacrificing spatial or spectral resolution. Second, the magnification and tilt of an individual grating controls the position of the dispersed image with respect to the optical axis, so that the spectral resolution is not constrained by the payload dimensions. Third, the radial symmetry of the design places the cameras closer together, resulting in a more compact instrument. Furthermore, by arranging the detectors around the optical axis, more dispersed grating orders can be populated; up to eight gratings can be arrayed around the ESIS primary mirror (up to six with the current optical table). This contrasts with the three image orders available in the planar symmetry of MOSES. Taken together, these three design features make ESIS more compact than MOSES (§ 2.1 item 1), improve spectral resolution (item 2) and allow the collection of more projections to better constrain the interpretation of the data (item 3).

The ESIS gratings are arranged in a segmented array, clocked in 45° increments, so that there are four distinct dispersion planes. This greatly aids in reconstructing spectral line profiles since the dispersion space of ESIS occupies a 3D volume rather than a 2D plane as with MOSES. For ESIS, there is always a dispersion plane within 22.5° of the normal to any loop-like feature in the solar atmosphere. As discussed in Section 2.1, a nearly perpendicular dispersion plane allows a filamentary structure to serve like a spectrographic slit, resulting in a clear presentation of the spectrum. This feature addresses § 2.1 item 4.

Rather than forming images at three spectral orders from a single grating, each ESIS imaging channel has a dedicated grating. Aberrations are controlled by optimizing the grating design to form images in first order, over a narrow range of ray deviation angles. This design controls aberration well enough to allow pixel-limited imaging, avoiding the PSF mismatch problems inherent to the MOSES design (§ 2.1 item 5). In its flight configuration with gratings optimized around a 629.732 Å wavelength, the instrument cannot be aligned and focused in visible light like MOSES. Visible gratings and a special alignment transfer procedure (§ **??**) are used to align and focus ESIS.

The ESIS design also includes an octagonal field stop placed at prime focus. This confers two advantages. First, the field stop fully defines the instrument FOV, so that ESIS is not susceptible to the spectral confusion observed in MOSES data (§ 2.1 item 6). Second, each spectral line image observed by ESIS is bordered by the outline of the field stop (e.g. § **??**). This aids the inversion process since outside of this sharp edge the intensity is zero for any look angle through an ESIS data cube. The size and octagonal shape of the field stop are defined by the requirement that all CCDs must see the entire FOV from edge to edge, while leaving a small margin for alignment.

Lastly, in contrast to MOSES, ESIS employs frame transfer CCDs to make optimum use of our five minutes of observing time. The ESIS design is shutterless, so

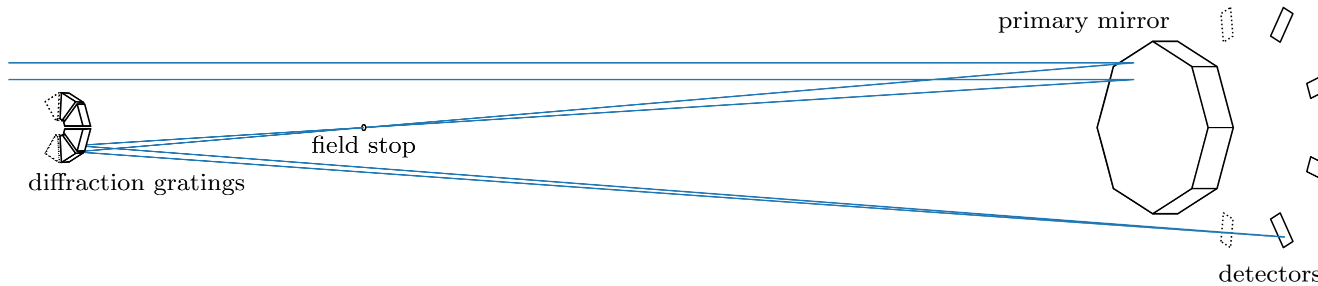

Figure 2. The ESIS optical layout. Dotted lines indicate the positions of unpopulated channels. The blue lines represent the path of O V through the system. The ESIS instrument is a pseudo-Gregorian design. The secondary mirror is replaced by a segmented array of concave diffraction gratings. The field stop at prime focus defines instrument spatial/spectral FOV. CCDs are arrayed around the primary mirror, each associated with a particular grating. Eight grating positions are available in principle; only six fit within the volume of the rocket payload. Four channels are populated for the first flight.

that each detector is always integrating. The result is a 100 % duty cycle. The lack of downtime for readout also allows ESIS to operate at a fixed, rapid cadence of 10 s. Longer integration times can be achieved for faint features by exposure stacking (§ 2.1 item 8).

In summary, the ESIS concept addresses all the limitations of the MOSES design enumerated in § 2.1. The volume of the ESIS optical layout is smaller than MOSES by almost a factor of two, yet with a smaller PSF, improved spectral resolution, and faster exposure cadence. ESIS offers several features to improve the recovery of spectral information, including more channels, crossed dispersion planes, and a field stop.

3. SCIENCE OBJECTIVES

Early flights of MOSES demonstrated a working concept of simultaneous EUV imaging and spectroscopy. This concept adds a unique capability to the science that we can obtain from the EUV solar atmosphere. ESIS as designed improves upon the MOSES concept, as discussed in the previous section, and therefore improves our ability to accomplish our scientific objectives. In this section, we set forth the specific scientific objectives of the ESIS mission. It is from these objectives that we derived the quantitative science requirements (§ 3.3) that drove the ESIS design.

The ESIS mission was designed to achieve the following two overarching science goals: (1) observe magnetic reconnection in the TR, and (2) map the transfer of energy through the TR with emphasis on magnetohydrodynamic (MHD) waves. These objectives have significant overlap with the missions of IRIS (B. De Pontieu et al. 2014), the EUV Imaging Spectrograph (EIS) (J. L. Culhane et al. 2007) aboard Hinode, the EUV Normal-incidence Spectrometer (EUNIS) (J. W. Brosius et al. 2007, 2014), and a long history of FUV and EUV slit spectrographs. The ESIS instrument, however, can obtain both spatial and spectral information co-temporally. This allows us to resolve complicated morphologies of compact TR reconnection events (as was done with MOSES (J. L. Fox 2011; T. L. Rust 2017; H. T. Courrier & C. C. Kankelborg 2018)) and observe signatures of MHD waves over a large portion of the solar disk. Therefore, in support of goal 1, we will use ESIS to map flows as a function of time and space in multiple TR reconnection events. To achieve goal 2, we will cross-correlate the evolution at multiple temperatures in the TR to map the vertical transport of energy over a wide FOV.

3.1. *Magnetic Reconnection Events*

Magnetic reconnection describes the re-arrangement of the magnetic topology wherein magnetic energy is converted to kinetic energy resulting in the acceleration of plasma particles. Reconnection is implicated in many dynamic, high-energy solar events. Solar flares are a well-studied example (e.g. E. R. Priest & T. G. Forbes (2002) and the references therein); however, we have little hope of pointing in the right place at the right time to observe a significant flare event in a rocket flight lasting only five minutes. Instead, we will search for signatures of magnetic reconnection in TR spectral lines. A particular signature of reconnection in the TR is the explosive energy release by ubiquitous, small-scale events. These explosive events (EEs) are characterized as spatially compact (≈1.5 Mm length (K. P. Dere 1994)) line broadenings on the order of 100 km s^{-1} (K. P. Dere et al. 1991). They are observed across a range of TR emission lines that span temperatures of 20 000–250 000 K (C II–O V) (D. Moses et al. 1994). The typical lifetime of an EE is 60–90 s (D. Moses et al. 1994; K. P. Dere 1994; K. P. Dere et al. 1991). Due to their location near quiet Sun magnetic network elements, and the presence of supersonic flows near the Alfvén speed, K. P. Dere et al. (1991) first suggested that EEs may result from the model of fast Petschek (H. E. Petschek 1964) reconnection.

The spectral line profile of EEs may indicate the type of reconnection that is occurring in the TR (e.g. T. L. Rust (2017)). For example, the Petschek model of reconnection predicts a 'bi-directional jet' line profile with highly Doppler-shifted wings, but little emission from the line core (D. E. Innes & G. Tóth 1999). D. E. Innes et al. (2015) developed a reconnection model resulting from a plasmoid instability (A. Bhattacharjee et al. 2009). In contrast to the bi-directional jet, this modeled line profile has bright core emission and broad wings. Both types of profile are seen in slit spectrograph data (e.g., D. E. Innes et al. (1997, 2015), and the references therein); however, MOSES observed EEs with more complicated morphologies than either of these two models suggest (J. L. Fox et al. 2010; T. L. Rust 2017). It is unclear whether the differing observations are a function of wavelength and temperature, a result of a limited number of observations, or a consequence of the morphology of the event being difficult to ascertain from slit spectrograph data.

ESIS will observe magnetic reconnection in the context of EEs, by extending the technique pioneered by MOSES to additional TR lines. Explosive events are well suited to sounding rocket observations; a significant portion of their temporal evolution can be captured in >150 s (e.g. the analysis by T. L. Rust (2017)) and they are sufficiently common to provide a statistically meaningful sample in a 5-minute rocket flight (e.g., K. P. Dere et al. (1989, 1991)). In similarity with MOSES, we sought a TR line for ESIS that is bright and well enough isolated from neighboring emission lines so as to be easily distinguished.

3.2. *Energy Transfer*

Tracking the mass and energy flow through the solar atmosphere is a long-standing goal in solar physics. Bulk mass flow is evidenced by Doppler shifts or skewness in spectral lines. However, the observed non-thermal broadening of TR spectral lines may result from a variety of physical processes, including MHD waves (B. De Pontieu et al. 2015, 2007), high-speed evaporative up-flows (e.g. nanoflares, S. Patsourakos & J. A. Klimchuk (2006)), turbulence, and other sources (e.g. J. T. Mariska (1992)). This is a broad topic which ESIS can address in many ways. Here we will focus on a single application; ESIS will search for sources of Alfvén waves in the solar atmosphere by observing Doppler shifts and line broadening as the spectroscopic signature of these waves. The spectroscopic signature of Alfvén waves is most obvious near the limb of the solar disk, since the motion of the ions is transverse with respect to the propagation direction. Pointing near the limb may make it more difficult to observe distinct EEs, described in Section 3.1, since the average width of TR lines often increases near the limb of the Sun (T. R. Ayres 2021), thus decreasing the signal-to-noise ratio (SNR) of single events.

Alfvén waves in coronal holes are observed to carry an energy flux of $7 \times 10^5\,\mathrm{erg/cm^2/s}$, enough to energize the fast solar wind (M. Hahn et al. 2012; M. Hahn & D. W. Savin 2013). The source and frequency spectrum of these waves is unknown. Here, we hypothesize that MHD waves are similarly ubiquitous in quiet Sun and active regions, and play an important role in the energization of the quiescent corona.

The magnitude of non-thermal broadening of optically thin spectral lines is a direct measure of the wave amplitude (D. Banerjee et al. 2009; M. Hahn et al. 2012; M. Hahn & D. W. Savin 2013). We may estimate a lower limit on the non-thermal velocity to be observed as follows. We assume that the magnetic field is constant for small changes in scale height in the TR and that line-of-sight effects are negligible for observations sufficiently far from disk center. Since the solar wind is not accelerated to an appreciable fraction of the Alfvén wave velocity at altitudes $R \leq 1.15R_\odot$ (S. R. Cranmer & A. A. van Ballegooijen 2005), the wave amplitude, v_{nt}, depends only weakly on electron density, n_e, so that $v_{nt} \propto n_e^{-1/4}$ (M. Hahn & D. W. Savin 2013; T. G. Moran 2001). Assuming pressure balance between the low corona and transition zone, we may infer non-thermal velocities in the TR by scaling according to the temperature drop, $v_{nt} \propto T^{1/4}$. The measured non-thermal velocity of $24\,\mathrm{km\,s^{-1}}$ for Si VIII (J. G. Doyle et al. 1998) (0.8 MK (T. G. Moran 2003)) near the limb should, neglecting damping, correspond to velocities of at least $18\,\mathrm{km\,s^{-1}}$ in the O V (0.25 MK) line. This is larger than the thermal width of $11\,\mathrm{km\,s^{-1}}$.

More recently, A. K. Srivastava et al. (2017) observed torsional Alfvén waves with amplitude $\sim 20\,\mathrm{km\,s^{-1}}$ and period ~ 30 s in the chromosphere. Modeling shows that these torsional waves can transfer a significant amount of energy to the corona (T. Kudoh & K. Shibata 1999). The torsional motion is observed as Doppler shifts when viewed from the side. The oscillation period is long enough to be well resolved but short enough to see ~ 10 cycles in a single rocket flight. An ESIS-like instrument is therefore well suited to observations of torsional Alfvén wave propagation over multiple heights in the TR.

By mapping Doppler velocities over a wide field of view in the TR, ESIS can address questions about both the origin of waves and whether they are able to propagate upward into the corona. Independent of the two propagation modes discussed above, there is a range of possible sources for Alfvén (and other MHD) waves in the solar atmosphere. Three potential scenarios are: (1) Waves originate in the chromosphere or below and propagate through the TR at a spatially uniform intensity; (2) Intense sources are localized in the TR, but fill only a fraction of the surface; and (3) Weak sources are localized in the TR, but cover the surface densely enough to appear like the first case. The resulting non-thermal widths for localized sources are significantly higher than the $\sim$18 km s^{-1} lower limit derived above. The concentration of non-thermal energy observed by ESIS will serve as an indicator of source density. Comparison of Doppler maps captured at different temperatures by ESIS will indicate whether a uniform source density originates in the chromosphere or below (scenario 1) or is associated with spatially distributed TR phenomena (scenario 3) such as explosive events, or macrospicules. Comparison with a wider selection of ground- and space-based imagery will allow us to determine whether intense, localized sources (scenario 2) are associated with converging or emerging magnetic bipoles, type II spicules, spicule bushes, or other sources beneath the TR. For these comparisons, we need only to localize, rather than resolve, wave sources. A spatial resolution of $\sim$2 Mm is sufficient to localize sources associated with magnetic flux tubes that are rooted in photospheric inter-granular network lanes (e.g. T. E. Berger et al. (1995)).

3.3. *Science Requirements*

ESIS will investigate two science targets: reconnection in explosive events and the transport of mass and energy through the transition region. The latter may take many forms, from MHD waves of various modes to EUV jets or macro-spicules. To fulfill these goals, ESIS will obtain simultaneous intensity, Doppler shift and line width images of the O V 629.732 Å line in the solar transition region (0.25 MK) at rapid cadence. The bright, optically thin O V emission line is well isolated except for the two coronal Mg X lines. These coronal lines can be viewed as contamination or as a bonus; we expect that with the four ESIS projections it is possible to separate the O V emission from that of Mg X. From the important temporal, spatial, and velocity scales referenced in Sections 3.1 and 3.2 we define the instrument requirements in Table 1 that are needed to meet our science goals.

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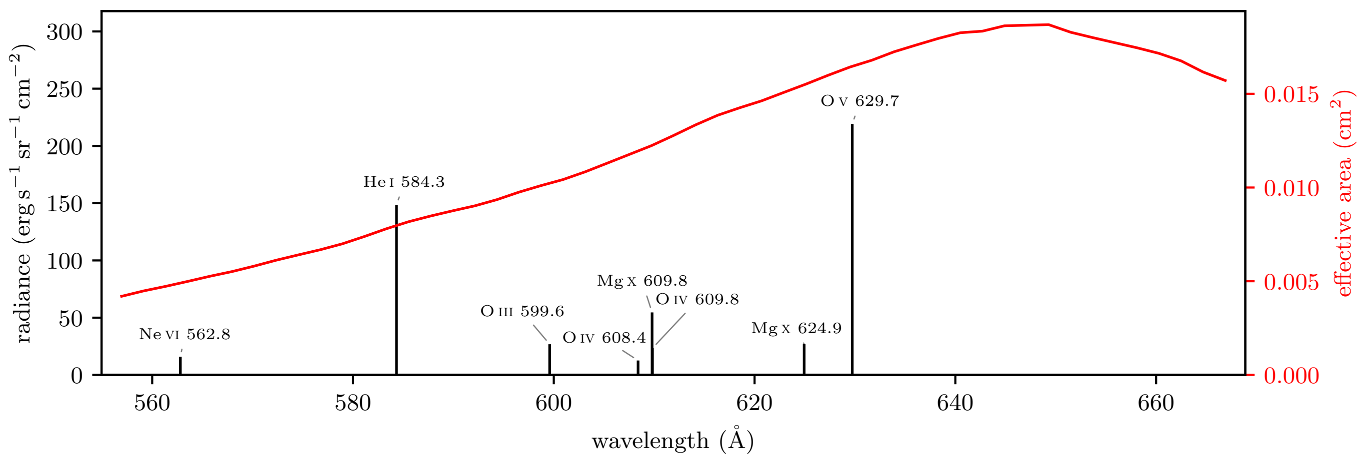

Figure 3. The eight brightest emission lines in the ESIS passband, and the effective area of a single channel beneath them. The lines are computed from version 11.0.2 of the CHIANTI atomic database, using the J. T. Schmelz et al. (2012) abundances of `sun_coronal_2012_schmelz_ext`, the quiet Sun differential emission measure (DEM) file `quiet_sun.dem`, and $n_e T = 1 \times 10^{15}$ K cm^{-3}.

Table 1. ESIS instrument requirements and capabilties. Note that MTF exceeds the Rayleigh criterion of 0.109.

Parameter	Requirement	Science Driver	Capabilities
Spectral line	O V 629.732 Å	EEs	O V, Mg X, He I, Figure 3
Spectral sampling	18 km s^{-1}	Broadening from MHD waves	17.5 km s^{-1} pix^{-1}, Table ??
Spatial resolution	2.1″(1.5 Mm)	EEs	??, Table ??
SNR	17.3 (CH)	MHD waves in CH	?? (12 × 10 s exp.), Table ??
Cadence	15 s	Torsional waves	10 s eff., Section ??
Observing time	150 s	EEs	306.8 s, Section ??
FOV diameter	10′	Span QS, AR, and limb	11.5′, Table ??

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